

Automating with SpiderFoot

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Before getting into the specific functions of Spiderfoot, it's important to first know that Spiderfoot is an OSINT framework. OSINT (Open Source Intelligence) refers to data that can be obtained from public sources, such as IP addresses, domain names, websites, and other social networks. Spiderfoot is created in Python and hosted on Github. The best thing about Spiderfoot is that it is open source, so anyone can help make it better. SpiderFoot can be used for a variety of tasks, including reconnaissance, information collection, and security analysis, making it an important tool in penetration testing and threat detection. SpiderFoot is a powerful cybersecurity tool that may be used both offensively and defensively.

Hardware

Downloading and Installation

For Spiderfoot, the downloading and installation process is very simple due to it being available on Github. Under the source code and attributes for Spiderfoot, Micallef lists an MIT license (Micallef) which states that any person may use the software without restrictions as long as the person is abiding by the rules of the program that runs the software. In the case for our example, Spiderfoot is being run in Kali Linux, therefore, the usage of Spiderfoot is at the discretion of Kali Linux terms of service (Micallef).

Spiderfoot is a public repository available for installation from github. The contents of the program and downloading instructions are available on Github by the github account name smicallef. The download and installation process does not require any prior downloading software in Kali Linux. The directions are explicit about the download process and include cloning the files directly from the git repository (*SpiderFoot – Automate...*).

Although Spiderfoot is mostly recommended for usage in Kali Linux, in the event that the user would like to utilize this program in Mac or Windows, prior installation may be needed. For Windows, not all windows devices have python installed. To download python onto Windows, a user can search the program on a traditional Microsoft Store application. For Mac, the user will need to install a program called brew. Brew is used to download the git repository. Once installed, the steps for Kali, Windows, and Mac all follow the same download and installation process.

Function

The purpose of spiderfoot is reconnaissance. With the active and passive scanning options on the Spiderfoot framework, Spiderfoot is able to obtain sufficient information and data about the target host (Intel471, n.d.). Spiderfoot gathers information using a variety of modules. Several scanning options and modules are provided to set and scan the target host within the Spiderfoot architecture. Spiderfoot can perform nearly all of the tasks you want for reconnaissance, depending on your needs. Each module in Spiderfoot creates events pertaining to data objects, which are subsequently received and watched over by other modules. With more than 100 modules, SpiderFoot enables everyone with an interest in security, from novices to experts, to comprehend their security perimeter (Jelen S, 2019)

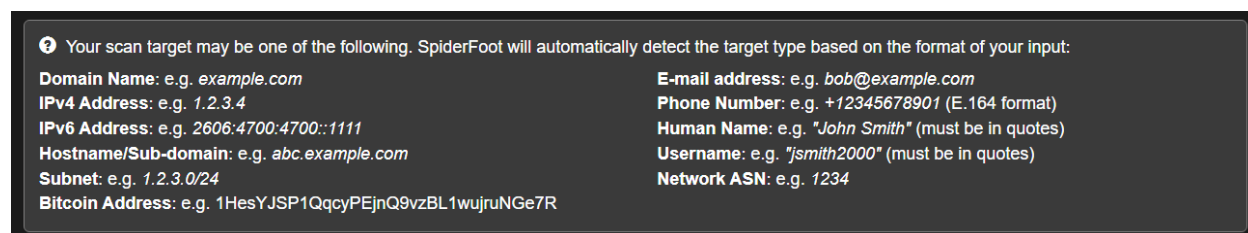
As mentioned earlier, Spiderfoot functions as a scanner for both passive and active target scanning. A Spiderfoot scan will yield a wealth of data and information about your target, including information on potential vulnerabilities, data breaches, and other sensitive information that may be used for threat intelligence, penetration testing, or red team exercises. A variety of data formats can be used by Spiderfoot as a "seed target." A "seed target" in Spiderfoot can be any of the following: a domain name, IP address (IPv4 or IPv6), hostname, subnet, ASN, email

address, phone number, human name, and username (Intel471, n.d.). After spiderfoot has finished scanning, you can quickly review the results. Spiderfoot's graphical data display capability is one of its better capabilities (Hackers Arise, 2022).

Usage and Features

Spiderfoot operates on a server IP address that runs on the terminal. The dashboard is located on a port, 127.0.0.1:5001, that can be typed into any browser search. The searching can be filtered through three different options. By use case – which deciphers between all available data, digital footprint, investigation on blacklists, and passive searching –, by required data – types of website the data could be available on, whether malicious or not, –, and by module – exact website or resource that collects and has a user's data publicly available.

When introducing a new scan to the program, there are various types of data you can search about. Depending on how the data is formatted into the search bar determines the format type. A single address can appear in multiple types, a name can also be a username for example.



The data that is returned from the search varies depending on the specificity of the search and the availability of public data. A factor about Spiderfoot that is important is that Spiderfoot does not have access to all data available on the internet. For example, a username from Instagram may not pop up due to the privacy and security policies Meta places on its applications. Another thing to note about the data from Spiderfoot is that all the data from Spiderfoot that is returned may not be relevant to the scan target. If you look at the common names available, it is very likely that there are multiple accounts attached to the same name but of different people.

Protection and Prevention

SpiderFoot can be used to reveal numerous pieces of information about a target and use that information to protect against attacks. At its basic capabilities, SpiderFoot can be used to check how vulnerable a network, IP address, or email account is to the world wide web (Intel471, n.d.). By simply entering a username into the scan for example, the user can view all of the accounts that specific username is connected to (Intel471, n.d.). This is helpful in determining how vulnerable a network is and informs the user about which vulnerabilities exist in order for them to be fixed (Intel471, n.d.). Noticing potential leaks of data is also a way SpiderFoot helps in securing a network, since it can track where specific pieces of information show up on the Internet (Intel471, n.d.). SpiderFoot is crucial in understanding the threats that pose a system and in helping users become more aware of the vulnerabilities that plague a system.

Creation

History of Creation

SpiderFoot was created in 2005 by Australian programmer Steve Micallef when he was 25 years old (Jelen, n.d.). The program was originally developed in C# with no intention of becoming an Open Source Intelligence tool; Micallef simply wanted to play around and learn C# (Jelen, n.d.). He did this while working IT at a mobile banking startup, balancing his work life and personal coding projects in Australia (Jelen, n.d.). Even though Micallef released Spiderfoot as open source, he shelved the project because it wasn't very popular (Jelen, n.d.). It wasn't until 7 years later in 2012 that Micallef returned to the SpiderFoot program and rewrote everything using Python (Jelen, n.d.). His reasoning behind this decision was simply that he wanted to learn how to use Python, and that version is the current SpiderFoot that is used today (Jelen, n.d.).

Most recently in 2022, Intel 471 acquired Spiderfoot in a deal and added it to their growing cybersecurity and cyber threat intelligence network (Intel471, 2022). Intel 471 is a privately owned data collection and threat analysis company that provides vulnerability assessment for the government and other companies (Intel471, 2022). According to Intel 471, the acquisition of Spiderfoot will greatly expand their scope of threat analysis and utilize Spiderfoot's capabilities to increase the strength of Intel 471's other cybersecurity programs. Also part of the deal was that Spiderfoot founder Steve Micallef would join the Intel 471 team as Vice President of Attack Surface Technology (Intel471, 2022). The acquisition broadens the scope of Spiderfoot and allows for it to be applied to numerous situations and environments (Intel471, 2022).

Application

As mentioned earlier, Spiderfoot is a free and open-source program hosted on Github, that can be used through Kali Linux. Python code was used to create the framework for this utility. To use this framework, your Kali Linux operating system has to have Python installed (GeeksforGeeks, 2018). The tool can use an array of resources to collect data, including popular services like Google, SANS, SHODAN, RIPE, Whois, PasteBin, and others. It is designed to extract the maximum amount of data possible, passing each piece of information to modules that could find it relevant in order to maximize the extraction of useful information. Also, in order to identify known dangerous entities. Spiderfoot integrates with sources and feeds of threat intelligence (Intel471, n.d.). This is extremely helpful in identifying cyber threats. Lastly, SpiderFoot scan results can be exported for reporting, sharing, and additional analysis in HTML, XML, CSV, and JSON forms (Intel471, 2022). Spiderfoots effectiveness ensures that it is a great tool for any and all cybersecurity projects and applications.

Conclusion

As OSINT tools are rapidly becoming a key science and skill for investigators and data scientists, Spiderfoot can save countless hours of labor when working without the need for individual tools. Although the data that Spiderfoot uncovers is meant to serve as a starting point, it can still be used to reveal numerous pieces of information about a target for offensive or defensive purposes. Its history shows how creative the process of program creation can be, and how it evolves to become something much more than what was originally intended. A SpiderFoot scan can reveal numerous aspects of a domain, email, or username, which can help security experts detail the vulnerabilities that lie in an organization's accounts. SpiderFoot is a powerful reconnaissance tool that can reveal the vulnerabilities in a system in order to better understand the threats that endanger a system.

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